

Class 12: Genome Informatics

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Data Import

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes

```
dataset <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6"
expr <- read.table(url(dataset))
```

```
table(expr$geno)
```

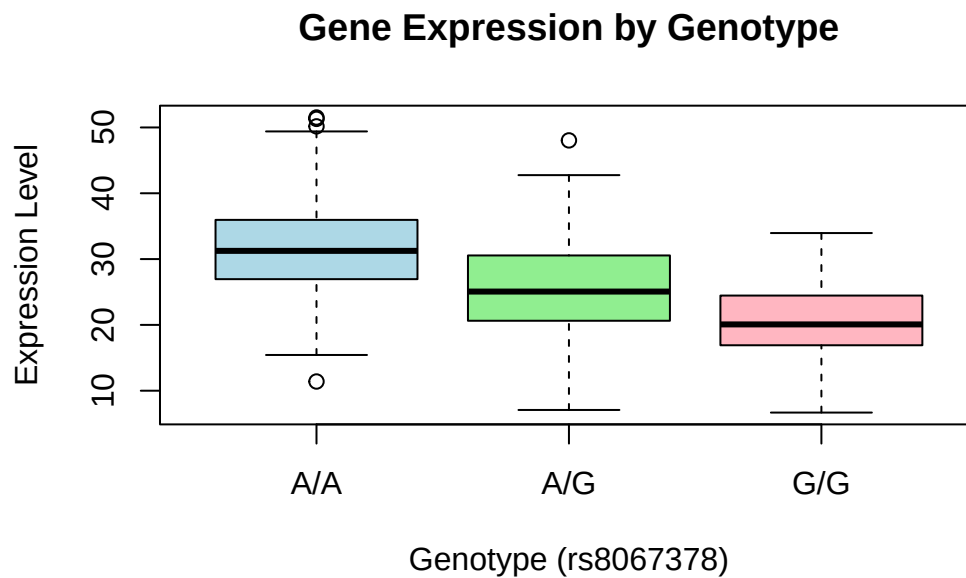
```
A/A A/G G/G
108 233 121
```

```
tapply(expr$exp, expr$geno, median)
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
p <- boxplot(exp ~ geno, data=expr,
             main="Gene Expression by Genotype",
             xlab="Genotype (rs8067378)",
             ylab="Expression Level",
             col=c("lightblue", "lightgreen", "lightpink"))
```



From the plot, we can see that the A/A genotype has the highest median at 31.25 and the G/G genotype has the lowest at 20.07. Yes, the SNP rs8067378 affects the expression of ORMDL3, as seen from the boxplots.

```
p$stats
```

	[,1]	[,2]	[,3]
[1,]	15.42908	7.07505	6.67482
[2,]	26.95022	20.62572	16.90256
[3,]	31.24847	25.06486	20.07363
[4,]	35.95503	30.55183	24.45672
[5,]	49.39612	42.75662	33.95602